

NELLIS AFB, NV, USA

WMO: 746140

Lat: 36.250N

Lon: 115.033W

Elev: 1870

StdP: 13.73

Time Zone: -8.00 (NAP)

Period: 96-19

WBAN: 23112

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	30.0	32.5	-1.8	5.4	50.4	2.5	6.7	52.0	28.6	50.1	26.1	51.6	3.2	40	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	109.2	66.7	107.1	66.2	104.9	65.6	72.5	94.0	71.4	93.9	70.5	94.2	9.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	104.6	80.9	63.9	95.4	83.9	61.6	87.7	86.3	37.3	92.7	36.3	93.6	35.4	94.1	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.5	24.4	20.6	25.5	112.8	2.6	2.2	23.6	114.4	22.1	115.7	20.7	117.0	18.8	118.6		
(5)			20.3	74.3	2.2	1.0	18.7	75.0	17.5	75.6	16.3	76.2	14.7	76.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.2	48.7	52.2	60.6	67.1	76.2	87.5	92.8	90.3	82.5	68.6	56.6	46.6		
	DBStd	17.01	5.38	6.44	6.16	6.77	7.32	5.84	4.27	4.36	6.23	6.92	6.74	5.42		
	HDD50	286	87	46	2	0	0	0	0	0	0	1	17	133		
	HDD65	1962	506	358	164	55	8	0	0	0	0	39	261	571		
	CDD50	7309	46	109	330	514	813	1125	1327	1250	975	579	214	27		
	CDD65	3509	0	1	27	119	355	675	862	785	525	152	8	0		
	CDH74	54291	0	24	378	1458	4680	10964	14761	12866	7339	1675	146	0		
	CDH80	34385	0	0	68	515	2417	7273	10428	8715	4350	607	12	0		
(14) Wind	WSAvg	7.7	5.8	7.3	8.5	10.0	9.7	9.1	8.1	7.5	7.2	7.0	5.7	6.0		
(15) Precipitation	PrecAvg	6.9	1.0	1.4	0.7	0.4	0.1	0.1	0.6	0.5	0.4	0.6	0.4	0.8		
	PrecMax	13.1	4.0	3.8	4.7	1.7	0.4	0.8	2.2	1.9	2.4	1.9	2.2	3.7		
	PrecMin	2.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0		
	PrecStd	2.6	1.1	1.1	0.9	0.4	0.1	0.2	0.4	0.4	0.6	0.6	0.5	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	77.3	86.1	95.0	102.0	111.6	113.2	109.4	105.0	95.4	81.7	67.8		
		MCWB	49.7	52.2	55.6	58.6	61.2	65.0	67.5	67.0	66.5	60.1	55.2	50.9		
	2%	DB	66.2	73.3	81.5	89.9	98.1	108.1	110.4	107.4	102.1	90.0	78.1	63.8		
		MCWB	47.6	50.7	54.4	57.0	59.7	64.7	67.5	66.6	65.2	59.5	53.5	48.4		
	5%	DB	63.5	69.4	78.3	85.9	95.2	105.4	108.0	105.4	99.7	86.3	75.0	60.8		
		MCWB	46.7	49.4	52.5	55.0	58.6	63.2	67.2	66.1	64.4	58.0	52.5	46.9		
	10%	DB	60.4	65.5	74.9	82.2	91.7	102.7	105.5	103.4	97.2	83.0	71.1	57.7		
		MCWB	45.3	48.3	51.0	53.8	57.7	62.2	66.7	66.1	63.5	56.6	51.0	45.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.6	55.4	57.2	59.8	62.5	69.4	74.0	73.8	71.3	66.3	57.8	55.3		
		MCDB	59.5	66.7	77.5	88.9	94.6	103.0	95.1	93.5	91.3	80.2	74.9	61.6		
	2%	WB	51.8	53.3	55.4	58.1	61.1	67.2	72.8	72.5	69.4	63.2	55.3	52.7		
		MCDB	58.8	65.8	76.6	85.0	93.0	101.6	95.0	93.4	92.9	80.0	74.4	58.8		
	5%	WB	50.0	51.7	53.8	56.4	59.8	65.4	71.8	71.4	68.2	61.0	53.4	50.4		
		MCDB	58.6	63.2	75.0	81.5	91.0	99.0	94.5	93.2	91.2	79.2	72.0	57.6		
	10%	WB	47.8	50.1	52.1	54.8	58.8	63.6	70.9	70.3	66.7	59.2	51.6	47.8		
		MCDB	57.3	63.2	72.4	79.2	89.0	99.2	94.7	92.9	90.1	78.0	69.6	55.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	20.8	22.9	23.5	23.0	25.3	22.8	23.6	24.3	23.7	24.4	19.8		
		MCDBR	26.6	27.8	27.5	29.8	28.6	28.8	26.6	26.7	26.7	27.6	28.6	24.0		
		MCWBR	14.1	13.7	12.0	13.0	11.2	10.8	9.4	10.0	10.6	12.3	14.5	13.4		
	5% WB	MCDBR	19.2	21.5	25.4	25.6	25.4	25.1	19.9	20.4	21.5	21.7	25.4	18.2		
		MCWBR	10.6	11.2	11.4	11.3	9.8	9.9	6.2	7.8	9.5	9.7	13.1	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.272	0.286	0.309	0.327	0.323	0.365	0.355	0.319	0.293	0.270	0.266		
	taud		2.613	2.615	2.551	2.464	2.438	2.458	2.431	2.450	2.530	2.587	2.620	2.622		
	Ebn at Noon		302	312	314	309	303	302	289	291	298	299	298	295		
	Edh at Noon		23	26	30	35	36	36	36	35	30	26	23	21		
(44) All-Sky Solar Radiation	RadAvg		968	1271	1750	2197	2493	2707	2392	2220	1958	1465	1091	855		
	RadStd		76	110	83	94	121	109	121	109	80	87	44	64		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	+0.88	N/A	-2.83	N/A	N/A	N/A	N/A	-211	+340	+146

Nomenclature: See separate page